Misc Exercises

September 19, 2023

Proof of the Similarity of Parabolas

We consider the vertex form of any quadratic function, $y = a(x-h)^2 + k$. Changing x and y does not change the steepnees of the curve of the parabola, but only the vertex (a fact that we decided should wait to be proven until we at least learn a bit about complex numbers), and hence, the actual curvature is given by $y = ax^2$. Now, consider any other a, let it be m; hence $m = al$ for some l. Well, we multiply both x and y in the above equation by l, thus recieving:
$yl = a(xl)^2$
$\rightarrow yl = ax^2l^2$
$\rightarrow y = alx^2$. Thus, we have effectively gone from every quadratic equation in vertex form by scaling the x and y coordinates, which proves that all quadratic equations are similar.